### 2.1.1 Atomic structure and isotopes

| | Learning outcomes | Additional guidance |
|---|---|---|
| | *Learners should be able to demonstrate and apply their knowledge and understanding of:* | |
| **Atomic structure and isotopes** | | |
| **(a)** | isotopes as atoms of the same element with different numbers of neutrons and different masses | |
| **(b)** | atomic structure in terms of the numbers of protons, neutrons and electrons for atoms and ions, given the atomic number, mass number and any ionic charge | HSW1 Different models for atomic structure can be used to explain different phenomena, e.g. the Bohr model explains periodic properties.<br><br>HSW7 The changing accepted models of atomic structure over time. The use of evidence to accept or reject particular models. |

**Relative mass**

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|---|---|---|
| **(c)** | explanation of the terms *relative isotopic mass* (mass compared with 1/12th mass of carbon-12) and *relative atomic mass* (weighted mean mass compared with 1/12th mass of carbon-12), based on the mass of a $^{12}C$ atom, the standard for atomic masses | Definitions required. |
| **(d)** | use of mass spectrometry in:<br>**(i)** the determination of relative isotopic masses and relative abundances of the isotope,<br>(ii) calculation of the relative atomic mass of an element from the relative abundances of its isotopes | *M0.2, M1.2, M3.1*<br>Knowledge of the mass spectrometer **not** required.<br>Limited to ions with single charges. |
| **(e)** | use of the terms *relative molecular mass*, $M_r$, and *relative formula mass* and their calculation from relative atomic masses. | For simple molecules, the term *relative molecular mass* will be used.<br>For compounds with giant structures, the term *relative formula mass* will be used.<br>Definitions of relative molecular mass and relative formula mass will **not** be required. |